PROPER HYPERCOVERINGS OF TOPOLOGICAL SPACES: A^{-1} ON SHEAVES IS FULLY FAITHFUL WITH ESSENTIAL IMAGE THE CARTESIAN SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the fully-faithful half is obtained by what the author calls 'a formal argument (chasing around the diagram)' (source line 5982) - the same glue step documented on cards L46151 and L46161, with all inputs (full faithfulness of h^{-1} and a_{qc}^{-1} with quasi-inverses) established before the sentence; the essential-image half is carried by the complete six-equality computation, each equality justified in-span; the displayed align block is reproduced verbatim.

Editorial note. The label lemma-descent-sheaves-for-proper-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The comparison diagram for the qc and Zariski simplicial sites. Same chapter, \label{lemma-compare-simplicial-objects}. Lines 5963-5965: the diagram h with h_{-1} = a and components h_n.
- Pullback along the map of locally compact spaces. Cohomology on Sites chapter, \label{sites-cohomology-lemma-describe-pullback-pi}. Lines 5966-5968: (h_{-1})^{-1} fully faithful with quasi-inverse h_{-1,*}, likewise componentwise.
- Description of the simplicial-site functor pair h^{-1}, h_*. Same chapter, \label{lemma-morphism-simplicial-sites} (card L46154 of this batch). Lines 5969-5970: componentwise full faithfulness lifts to h^{-1} fully faithful with quasi-inverse h_*.
- Proper surjective maps are qc coverings. Cohomology on Sites chapter, \label{sites-cohomology-lemma-proper-surjective-is-qc-covering}. Lines 5972-5975: U is a hypercovering of X in LC_qc.
- Simple sheaf descent for hypercoverings. Same chapter, \label{lemma-hypercovering-X-simple-descent-sheaves}. Lines 5976-5979: a_{qc}^{-1} fully faithful with essential image the cartesian sheaves on the total site.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Proper push-pull on locally compact spaces. Cohomology on Sites chapter, \label{sites-cohomology-lemma-push-pull-LC}. Lines 6020-6024: the third equality of the equalizer computation, using that a_0 and a_1 are proper.
- The equalizer description of a_* and of the total-site augmentation. Same chapter, \label{lemma-augmentation} and \label{lemma-augmentation-site}. Lines 6010-6018 and 6025-6026: the first and fifth equalities of the computation.

Source reference. Original reference 09W4: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 524–524. Prerequisite pointer invoked at proof line(s) 5959; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D70: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1141–1141. Prerequisite pointer invoked at proof line(s) 6035; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DAF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5910–5910. Prerequisite pointer invoked at proof line(s) 5965; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DA8: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5065–5065. Prerequisite pointer invoked at proof line(s) 5976; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D96: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1273–1273. Prerequisite pointer invoked at proof line(s) 5970; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 09A0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5012–5012. Prerequisite pointer invoked at proof line(s) 5973; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09X3: sources/stacks-chow/tags.txt, source lines 12516–12516. Prerequisite pointer invoked at proof line(s) 5966; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09X5: sources/stacks-chow/tags.txt, source lines 12518–12518. Prerequisite pointer invoked at proof line(s) 5975; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D92: sources/stacks-chow/tags.txt, source lines 16612–16612. Prerequisite pointer invoked at proof line(s) 6032; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let U be a simplicial object of LC and let $a : U \to X$ be an augmentation. If $a : U \to X$ gives a proper hypercovering of X, then*

$$a^{-1} : Sh(X) \to Sh(U_{Zar}) \quad \text{and} \quad a^{-1} : Ab(X) \to Ab(U_{Zar})$$

are fully faithful with essential image the cartesian sheaves and quasi-inverse given by a_*. Here $a : Sh(U_{Zar}) \to Sh(X)$ is as in Lemma 09W4.

Proof. We will prove the statement for sheaves of sets. It will be an almost formal consequence of results already established. Consider the diagram of Lemma 0DAF. By Cohomology on Sites, Lemma 09X3 the functor $(h_{-1})^{-1}$ is fully faithful with quasi-inverse $h_{-1,*}$. The same holds true for the components h_n of h. By the description of the functors h^{-1} and h_* of Lemma 0D96 we conclude that h^{-1} is fully faithful with quasi-inverse h_*. Observe that U is a hypercovering of X in LC_{qc} (as defined in Section 09A0) by Cohomology on Sites, Lemma 09X5. By Lemma 0DA8 we see that a_{qc}^{-1} is fully faithful with quasi-inverse $a_{qc,*}$ and with essential image the cartesian sheaves on $(LC_{qc}/U)_{total}$. A formal argument (chasing around the diagram) now shows that a^{-1} is fully faithful.

Finally, suppose that $\mathcal{G}$ is a cartesian sheaf on U_{Zar}. Then $h^{-1}\mathcal{G}$ is a cartesian sheaf on LC_{qc}/U. Hence $h^{-1}\mathcal{G} = a_{qc}^{-1}\mathcal{H}$ for some sheaf $\mathcal{H}$ on LC_{qc}/X. We compute

$$\begin{aligned}
(h_{-1})^{-1}(a_*\mathcal{G}) &= (h_{-1})^{-1}\mathrm{Eq}(\ a_{0,*}\mathcal{G}_0 \rightrightarrows a_{1,*}\mathcal{G}_1\ ) \\
&= \mathrm{Eq}(\ (h_{-1})^{-1}a_{0,*}\mathcal{G}_0 \rightrightarrows (h_{-1})^{-1}a_{1,*}\mathcal{G}_1\ ) \\
&= \mathrm{Eq}(\ a_{qc,0,*}h_0^{-1}\mathcal{G}_0 \rightrightarrows a_{qc,1,*}h_1^{-1}\mathcal{G}_1\ ) \\
&= \mathrm{Eq}(\ a_{qc,0,*}a_{qc,0}^{-1}\mathcal{H} \rightrightarrows a_{qc,1,*}a_{qc,1}^{-1}\mathcal{H}\ ) \\
&= a_{qc,*}a_{qc}^{-1}\mathcal{H} \\
&= \mathcal{H}
\end{aligned}$$

Here the first equality follows from Lemma 09W4, the second equality follows as $(h_{-1})^{-1}$ is an exact functor, the third equality follows from Cohomology on Sites, Lemma 0D92 (here we use that $a_0 : U_0 \to X$ and $a_1 : U_1 \to X$ are proper), the fourth follows from $a_{qc}^{-1}\mathcal{H} = h^{-1}\mathcal{G}$, the fifth from Lemma 0D70, and the sixth we've seen above. Since $a_{qc}^{-1}\mathcal{H} = h^{-1}\mathcal{G}$ we deduce that $h^{-1}\mathcal{G} \cong h^{-1}a^{-1}a_*\mathcal{G}$ which ends the proof by fully faithfulness of h^{-1}. □